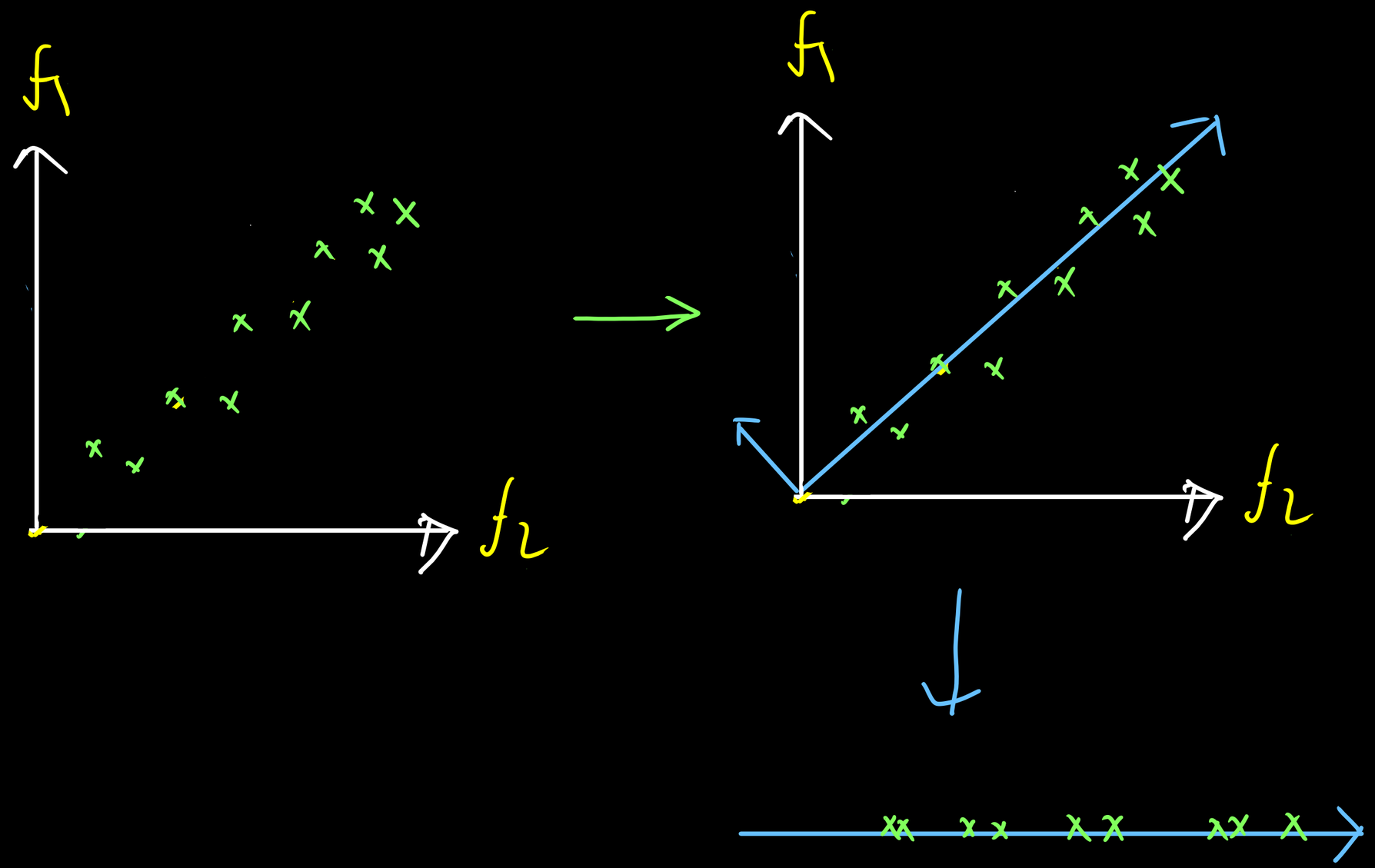
Agenda

1. Assumption of PCA
2. Limitation of PCA
3. t-SNE


f_1
f_2
f_1
f_2

Steps for PCA

1. Standardize the data ($\mu = 0$, $\sigma = 1$)
2. Transform m-dim. into m-dim in diff. coordinate system (Eigen Vector or Princ. Component)
   - Compute the covariance matrix $X^T X$.
   - Compute eigen value & eigen vector
   - Sort the eigen vector w.r.t eigen value
3. Take top-K eigen vector corresponding to top-K eigen value.
4. Project the data into top-K eigen vector.

Assumption of PCA

① PCA carries out linear transformation

→ Can PCA work with correlated data ?

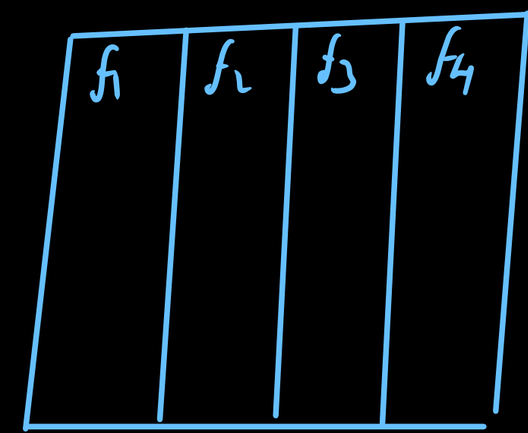
m-dim → m-dim diff coordinate (Eigen Vector) $\xrightarrow{\text{top-K}}$ K-dimen

↓ Correlated

↓ Uncorrelated

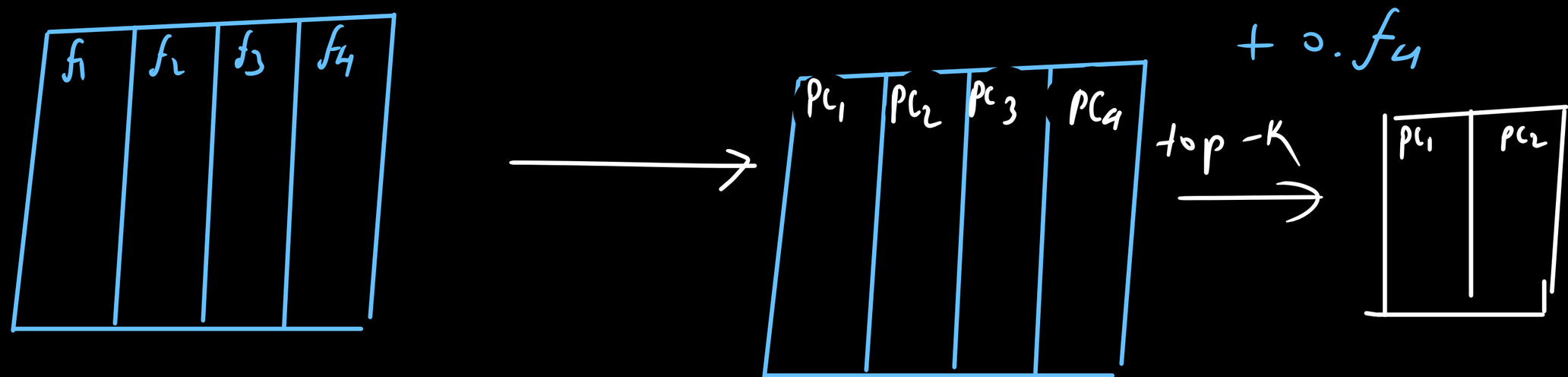
Limitation of PCA

1) Interpretibity

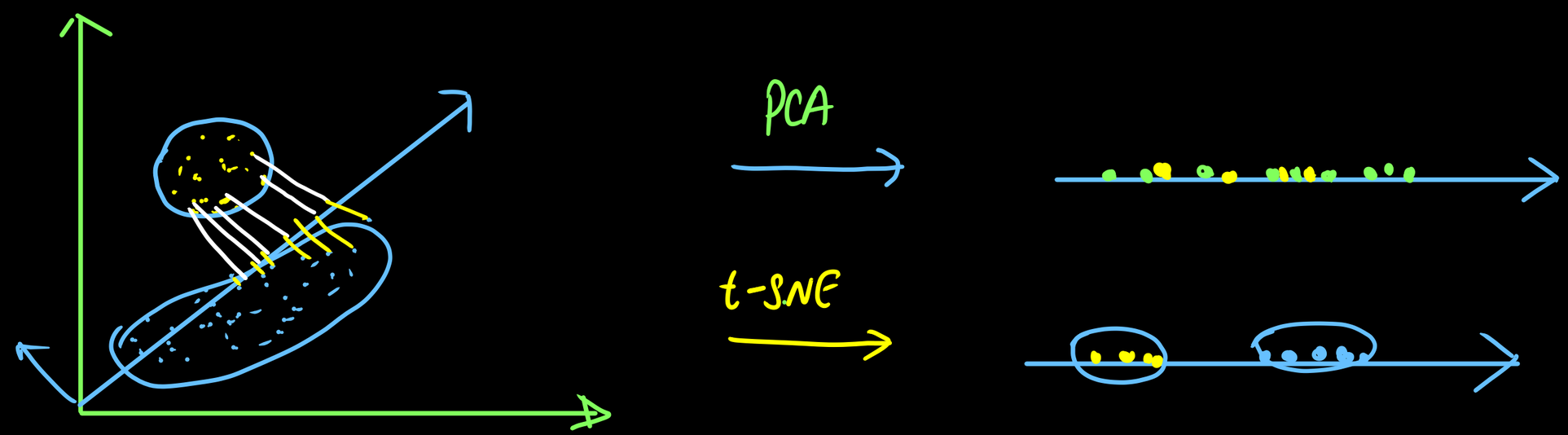


Linear

$$\hat{y} = 2f_1 + 0f_2 + 10f_3 + 0.f_4$$



3) PCA cannot preserve local structure

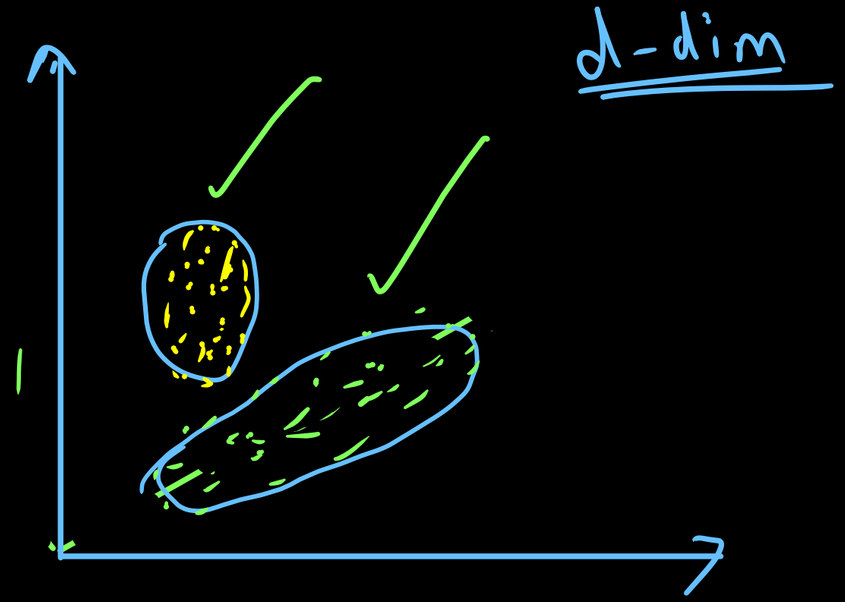


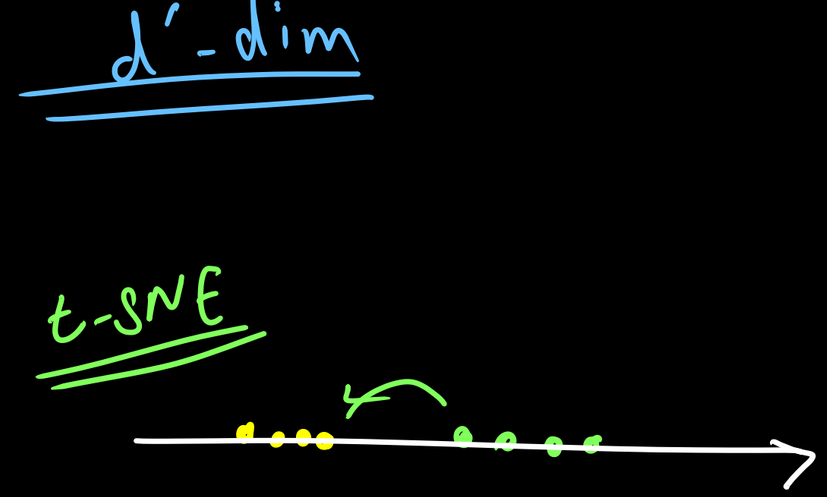
→ Break until 22:12 PM

t-SNE

Stochastic Neighbor Embedding.

→ Preserve local & Global Structure





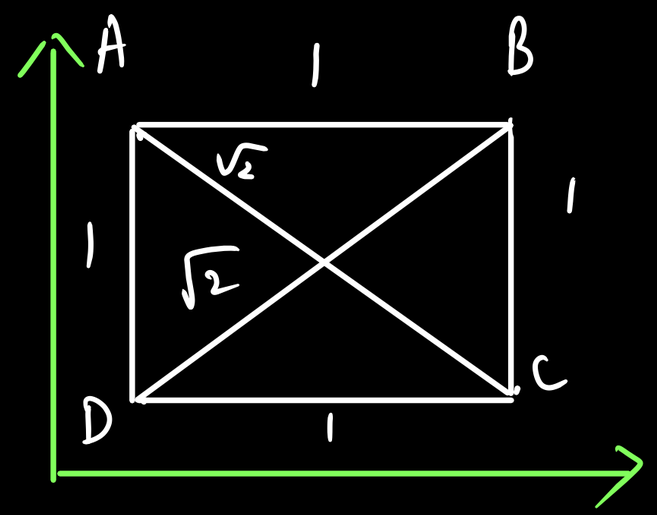
1) Points closer in higher dim will be closer in lower dim

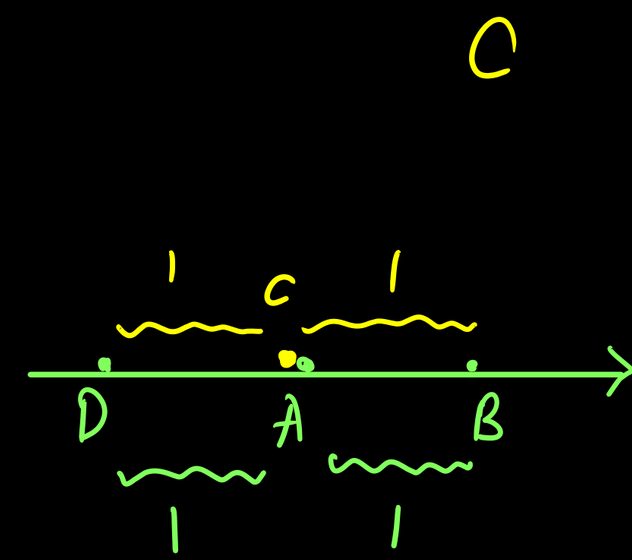
2) Points farther in higher dim will be farther in lower dim

↳ t-SNE will preserve local + global structure

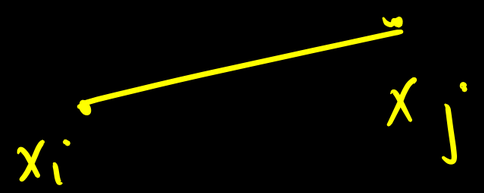
d-dim	d'-dim
$x_i \in R^d$	$y \in R^{d'}$
$x_i \longrightarrow x_j$	$y_i \longrightarrow y_j$
$x_j \in Neigh(x_i)$	$y_j \in Neighbor(y_i)$

→ Crowding problem



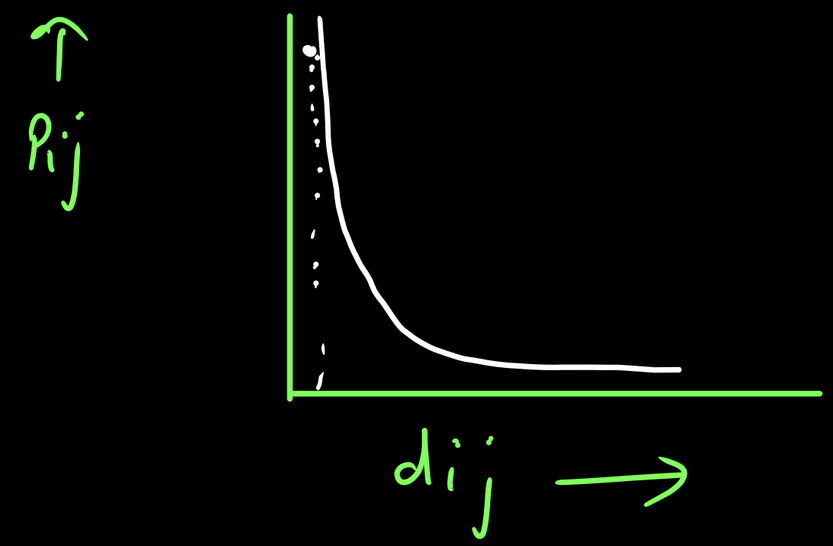


→ Due to crowding problem, we CANNOT preserve the distance. Hence we try to preserve Probability.



$$p_{ij} \propto \frac{1}{\|x_i - x_j\|_2}$$

$$\propto \boxed{\frac{1}{d_{ij}}}$$

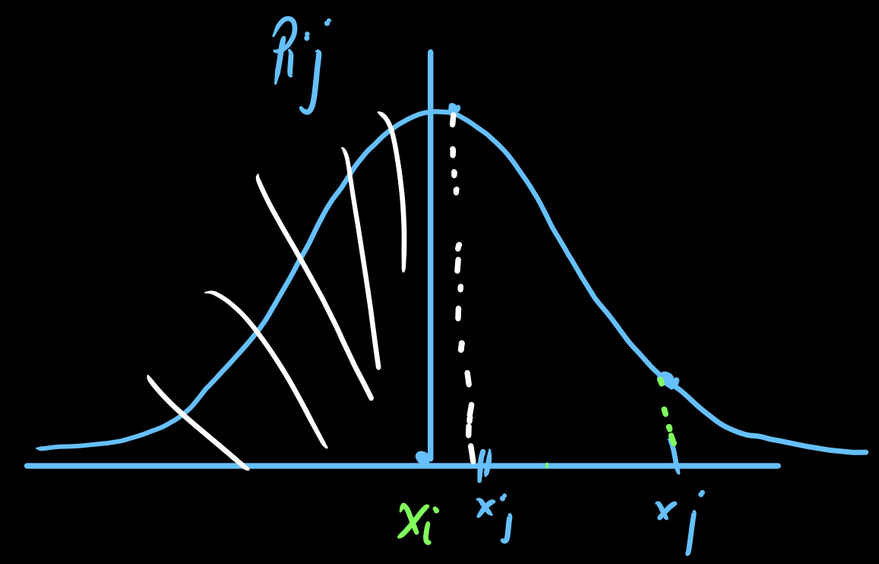


Problem

→ Steepness is very high

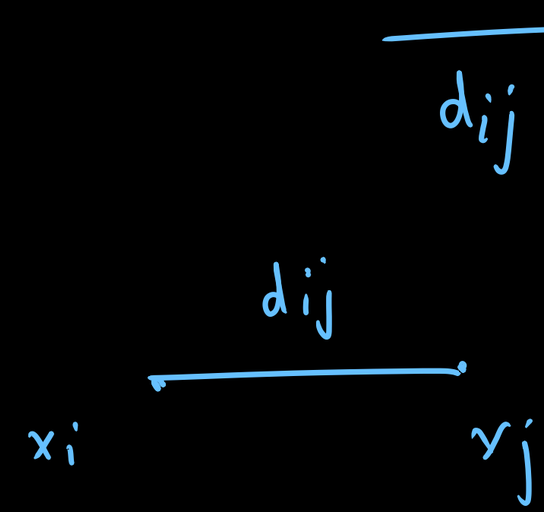
$$p_{ij} = e^{-\frac{\|x_i - x_j\|^2}{2\sigma_i^2}}$$

$$= \frac{1}{e^{d_{ij}^2/2\sigma_i^2}}$$

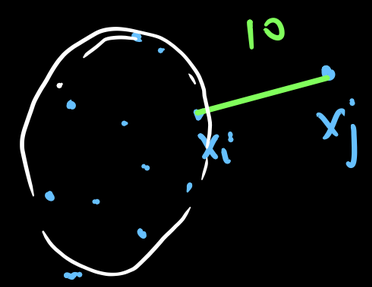


$$d_{ij} = \|x_i - x_j\|^2$$

$$\boxed{p_{ij} \propto \frac{1}{d_{ij}}}$$

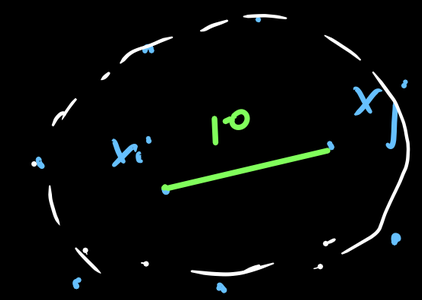


Case 1 → Dense



$$\boxed{p_{ij} = e^{-\frac{\|x_i - x_j\|^2}{2\sigma_i^2}}}$$

Case 2 → Sparse



2

$\sigma_i = 1$

$$p_{ij} = e^{-10^2/2\times 1}$$

$$= e^{-50} \rightarrow \text{Very low probab}$$

$\sigma_i = 3$

$$p_{ij} = e^{-10^2/2\times 3}$$

$$= e^{-16.66}$$

↳ Higher

$$\frac{X - \mu}{\sigma}$$

$\mu \rightarrow$ Test data

$\sigma \rightarrow$

Categorical data

↳ Embedding

↳ PCA